Qualification Pack

Automotive Welding Machine Technician

QP Code: ASC/Q3103

Version: 7.0

NSQF Level: 4

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Contents

ASC/Q3103: Automotive Welding Machine Technician

Brief Job Description

The individual is primarily involved in all robotic and manual welding operations performed in automotive manufacturing. They use various types of welding processes such as TIG, MIG, SMAW welding etc. The individual perform activities such as inspection of equipment condition, gauging, testing and inspection of welded work pieces.

Personal Attributes

The person should be patient, organised, team-oriented and have the ability to work for long hours in adverse conditions. They should be keen observers and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ASC/N9803: Organize work and resources (Manufacturing)
2. DGT/VSQ/N0102: Employability Skills (60 Hours)
3. ASC/N9805: Interpret engineering drawing
4. ASC/N3134: Perform robotic welding operations
5. ASC/N3136: Perform welding and post welding operations

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Metal Joining
Country	India
NSQF Level	4
Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7212.0302

Minimum Educational Qualification & Experience	10th Class (+1 year ITI) OR 10th Class with 2 Years of experience of relevant experience OR 11th Class OR Certificate-NSQF (Level 3 (Automotive Welding Machine Operator (Manual and Robotics) with 2 Years of experience of relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	25/02/2029
NSQC Approval Date	25/02/2026
Version	7.0
Reference code on NQR	QG-04-AM-05052-2025-V1-ASDC
NQR Version	1

ASC/N9803: Organize work and resources (Manufacturing)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising use of resources

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Health and hygiene
- Perform work as per quality standards
- Effective waste management practices
- Material/energy conservation practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace

PC2. follow safe working practices while dealing with hazards to ensure safety of self and others

PC3. carry out routine check of the machine for identifying potential hazards

PC4. use appropriate protective clothing/equipment for specific tasks and work

PC5. follow safety hazards and preventive techniques during fire drill

PC6. report any identified breaches in health, safety and security policies and procedures to the designated person

Health and hygiene

To be competent, the user/individual on the job must be able to:

PC7. ensure workstation and equipment are regularly clean and sanitized

PC8. clean hands with soap, alcohol-based sanitizer regularly

PC9. avoid contact with ill people and self-isolate in a similar situation

PC10. wear and dispose PPEs regularly and appropriately

PC11. report advanced hygiene and sanitation issues to appropriate authority

PC12. follow stress and anxiety management techniques

Perform work as per quality standards

To be competent, the user/individual on the job must be able to:

PC13. ensure that work is accomplished as per the requirements within the specified timeline

PC14. ensure team goals are given preference over individual goals

Effective waste management practices

To be competent, the user/individual on the job must be able to:

PC15. follow the fundamentals of 5S for waste management

PC16. segregate waste into different categories

PC17. follow processes specified for disposal of hazardous waste

PC18. identify recyclable, non-recyclable and hazardous waste

PC19. dispose non-recyclable, recyclable and reusable waste appropriately at identified location

Material/energy conservation practices

To be competent, the user/individual on the job must be able to:

PC20. identify ways to optimize usage of material in various tasks/activities/processes

PC21. check for spills/leakages in various tasks/activities/processes

PC22. plug spills/leakages and escalate to appropriate authority if unable to rectify

PC23. check if the equipment/machine is functioning normally before commencing work and rectify wherever required

PC24. report malfunctioning (fumes/ sparks/emission/vibration/noise) and lapse in maintenance of equipment

PC25. ensure electrical equipment and appliances are properly connected and turned off when not in use

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organisation procedures for health, safety and security, individual role and responsibilities in this context

KU2. the organisation's emergency procedures for different emergency situations and the importance of following the same

KU3. evacuation procedures for workers and visitors

KU4. how and when to report hazards as well as the limits of responsibility for dealing with hazards

KU5. potential hazards, risks and threats based on the nature of work

KU6. preventative and remedial actions to be taken in case of exposure to toxic material

KU7. various types of fire extinguisher

KU8. various types of safety signs and their meaning

KU9. appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.

KU10. relevant standards, procedures and policies related to 5S followed in the company

KU11. the various materials used and their storage norms

KU12. efficient utilisation of material and water

KU13. basics of electricity and prevalent energy efficient devices

KU14. common practices of conserving electricity

KU15. common sources and ways to minimize pollution

KU16. categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

KU17. usage of different colors of dustbins

KU18. waste management techniques

KU19. significance of greening

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read safety instructions/guidelines

GS2. modify work practices to improve them

GS3. ask for clarifications from superior about the job requirement

GS4. work with supervisors/team members to carry out work related tasks

GS5. complete tasks efficiently and accurately within stipulated time

GS6. inform/report to concerned person in case of any problem

GS7. make timely decisions for efficient utilization of resources

GS8. write reports such as accident report, in at least English/regional language

GS9. be punctual and utilize time efficiently

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Maintain safe and secure working environment	**11**	**5**	**-**	**7**
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	2	1	-	2
PC2. follow safe working practices while dealing with hazards to ensure safety of self and others	2	-	-	1
PC3. carry out routine check of the machine for identifying potential hazards	2	1	-	1
PC4. use appropriate protective clothing/equipment for specific tasks and work	2	1	-	1
PC5. follow safety hazards and preventive techniques during fire drill	2	1	-	1
PC6. report any identified breaches in health, safety and security policies and procedures to the designated person	1	1	-	1
Health and hygiene	**7**	**5**	**-**	**2**
PC7. ensure workstation and equipment are regularly clean and sanitized	2	2	-	1
PC8. clean hands with soap, alcohol-based sanitizer regularly	1	1	-	1
PC9. avoid contact with ill people and self-isolate in a similar situation	1	-	-	-
PC10. wear and dispose PPEs regularly and appropriately	1	-	-	-
PC11. report advanced hygiene and sanitation issues to appropriate authority	1	1	-	-
PC12. follow stress and anxiety management techniques	1	1	-	-
Perform work as per quality standards	**5**	**3**	**-**	**2**
PC13. ensure that work is accomplished as per the requirements within the specified timeline	2	2	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. ensure team goals are given preference over individual goals	3	1	-	1
Effective waste management practices	**15**	**10**	**-**	**4**
PC15. follow the fundamentals of 5S for waste management	3	2	-	1
PC16. segregate waste into different categories	2	1	-	-
PC17. follow processes specified for disposal of hazardous waste	2	2	-	1
PC18. identify recyclable, non-recyclable and hazardous waste	4	2	-	1
PC19. dispose non-recyclable, recyclable and reusable waste appropriately at identified location	4	3	-	1
Material/energy conservation practices	**12**	**7**	**-**	**5**
PC20. identify ways to optimize usage of material in various tasks/activities/processes	2	1	-	1
PC21. check for spills/leakages in various tasks/activities/processes	2	1	-	1
PC22. plug spills/leakages and escalate to appropriate authority if unable to rectify	2	1	-	-
PC23. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	2	2	-	1
PC24. report malfunctioning (fumes/ sparks/emission/vibration/noise) and lapse in maintenance of equipment	2	1	-	1
PC25. ensure electrical equipment and appliances are properly connected and turned off when not in use	2	1	-	1
NOS Total	**50**	**30**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9803
NOS Name	Organize work and resources (Manufacturing)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	3
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

ASC/N9805: Interpret engineering drawing

Description

This NOS unit is about reading and interpreting all concepts, symbols, methods, views, etc. of engineering drawing.

Scope

The scope covers the following :

- Interpret information from various views, projection, 2D and 3D shapes
- Identify drawing standards and symbols
- Modification and storage of drawing

Elements and Performance Criteria

Interpret information from various views, projection, 2D and 3D shapes

To be competent, the user/individual on the job must be able to:

PC1. interpret engineering drawing's uniqueness, dimensions and important features in 2D and 3D shapes

PC2. identify the difference between 2D and 3D shapes

PC3. explain difference between first angle projection and third angle projection in mechanical engineering drawing

PC4. interpret all the 3 axes (x, y and z axis) and geometrical shapes (cones, cylinder, sphere, cuboid, etc) on to a 2D and 3D projection

PC5. identify details of the machine component which are not clearly visible by interpreting section views

Identify drawing standards and symbols

To be competent, the user/individual on the job must be able to:

PC6. interpret Geometric Dimensioning and Tolerencing (GD&T) symbols in the drawings

PC7. interpret symbols of Radius, controlled radius, spherical radius, diameter, spherical diameter, square, counterbore, spotface, depth, countersink, “by”, maximum dimension, minimum dimension, reference, dimension origin etc

PC8. identify the sequence of operations which enables the selection and prioritization of the datums

PC9. read and interpret information from Tolerance Zone boundaries for part features in terms of shape and size

Modification and storage of drawing

To be competent, the user/individual on the job must be able to:

PC10. observe any modification, changes required in the drawing and communicate the same to the concerned team in the organization

PC11. store the drawings in an easily accessible place, avoiding damage from moisture, chemicals and fire

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant organisational standards such as work standard, Standard Operating Procedure, quality process, maintenance standards etc. followed in the company
KU2. importance of cycle-time and required output as per work order and work instructions
KU3. drawing standards used by the company
KU4. use of drawing tools such as scales, compass, types of pencils, CAD and CAM software etc.
KU5. the basics of engineering drawing, orthographic projection, isometric projection, GD&T etc.
KU6. importance of various projections, views, symbols and dimensions of drawing
KU7. use of geometric shapes like lines, angles, circles, etc for interpreting the drawing

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret workplace related drawing
GS2. communicate the changes and requirements to supervisor by using relevant drawing terms and nomenclature
GS3. attentively listen and comprehend the information given by the supervisor/team members
GS4. write in English/regional language
GS5. recognise problem in drawing and take suitable action
GS6. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Interpret information from various views, projection, 2D and 3D shapes	**21**	**11**	**-**	**10**
PC1. interpret engineering drawing's uniqueness, dimensions and important features in 2D and 3D shapes	5	3	-	2
PC2. identify the difference between 2D and 3D shapes	4	2	-	2
PC3. explain difference between first angle projection and third angle projection in mechanical engineering drawing	4	-	-	2
PC4. interpret all the 3 axes (x, y and z axis) and geometrical shapes (cones, cylinder, sphere, cuboid, etc) on to a 2D and 3D projection	5	3	-	2
PC5. identify details of the machine component which are not clearly visible by interpreting section views	3	3	-	2
Identify drawing standards and symbols	**23**	**15**	**-**	**8**
PC6. interpret Geometric Dimensioning and Tolerencing (GD&T) symbols in the drawings	6	4	-	2
PC7. interpret symbols of Radius, controlled radius, spherical radius, diameter, spherical diameter, square, counterbore, spotface, depth, countersink, “by”, maximum dimension, minimum dimension, reference, dimension origin etc	6	4	-	2
PC8. identify the sequence of operations which enables the selection and prioritization of the datums	5	3	-	2
PC9. read and interpret information from Tolerance Zone boundaries for part features in terms of shape and size	6	4	-	2
Modification and storage of drawing	**6**	**4**	**-**	**2**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. observe any modification, changes required in the drawing and communicate the same to the concerned team in the organization	3	2	-	1
PC11. store the drawings in an easily accessible place, avoiding damage from moisture, chemicals and fire	3	2	-	1
NOS Total	**50**	**30**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9805
NOS Name	Interpret engineering drawing
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	1
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

ASC/N3134: Perform robotic welding operations

Description

This NOS unit is about performing robotic welding activities and post-welding operations as per the given work order and the standards specified by the organization

Scope

The scope covers the following :

- Prepare for robotic welding operations
- Write robotic welding program
- Perform robotic welding
- Perform post-welding operations

Elements and Performance Criteria

Prepare for robotic welding operations

To be competent, the user/individual on the job must be able to:

PC1. identify the work to be done and product specifications by interpreting the engineering drawing,Welding Procedure Specification (WPS) and job orders

PC2. identify and arrange required tools, robotic arm, measuring instruments, accessories, consumables and input materials as per the requirements mentioned in WPS or drawing

PC3. check that material to be welded is as per the required specifications, quality and job requirement

PC4. plan the welding activities before starting the actual process as per WPS

PC5. set welding parameters like current, voltage, electrode size, material thickness, and joint type as per the SOP

PC6. follow appropriate safety practices as specified by organization during the work

Write robotic welding program

To be competent, the user/individual on the job must be able to:

PC7. identify robotic technology, welding robot features and specifications including robot language and welding process and parameter capabilities to be used in welding applications

PC8. plan robot welding program

PC9. confirm required operations for robot arm and torch, types of weld, any positions or welds requiring non-standard instructions to be performed using appropriate technical reference source

PC10. review robot file library for correct references to weld size and weld type

PC11. identify suitable program code for robotic technology and welding applications

PC12. write joint arm and welding moves using teach pendant or software

PC13. use software including canned cycles and sub-routines for programming welding operations

PC14. program weld procedure for the robot using short, interrupted and continuous welds

PC15. create command instructions for welding instructions

PC16. perform any additional programming required including instructions, weld data recording and lower-level programming

PC17. check the sequence of the program as per the process sheet

PC18. test and debug the program for weld accuracy, compatibility and efficiency requirements by conducting trial run of the robotic arm on test material

PC19. verify trial welds and program performance against required specifications

PC20. edit program to adjust welding operations based on trial outcomes

PC21. save final program and complete operation sheets and any required records according to enterprise procedures

Perform robotic welding

To be competent, the user/individual on the job must be able to:

PC22. install the work pieces and fixture on the apparatus and align them with the robotic arm as per the job requirements

PC23. set up robotic arm for functional operation in accordance with manufacturer's specifications

PC24. follow welding instructions, visualize weld paths, and check for potential defects during the welding process

PC25. perform welding by operating robotic arm and tack weld the joint at appropriate intervals in accordance with job plan, specifications, relevant welding standards and manufacturers'

PC26. monitor the welding process parameters (air pressure, electrode force, electrode distance, gas flow etc.) during the welding process

PC27. produce joints of the specified dimensional accuracy and required weld quality

PC28. measure the final welded piece and compare with the dimensions as prescribed in the WPS and engineering drawing

PC29. remove extra material, distortion etc. by using chipping hammers, grinders etc., from the welded piece

Perform post-welding operations

To be competent, the user/individual on the job must be able to:

PC30. visually inspect welds to ensure that they meet specified standards and regulations

PC31. inspect welds to identify defects like porosity, undercuts, improper penetration etc

PC32. conduct destructive and non-destructive tests on the work pieces

PC33. tag and store the right quality pieces by following organizational policies and procedures

PC34. report to the supervisor about any problems faced or anticipated during the complete process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant standards and procedures followed in the company

KU2. overview of robotic automation and its application in welding

KU3. welding specific equipment requirements for robotic welding

KU4. equipment and robotic arm settings

KU5. different current and voltage settings, gas flow rates, electrode feed and other variables to suit typical situations

KU6. material and equipment preparation

KU7. properties and characteristics of materials and consumables

KU8. impact of various welding parameters like voltage, current, gas flow rate, speed, pressure, torch angle, cycle time, electrode distance etc. on the quality and quantity of welding

KU9. safety features and requirements of welding robots

KU10. welding symbols and abbreviations

KU11. welding parameters as applied to welding robots

KU12. welding robot programming including the use of standard program codes and comments, and input controls for welding robot programs

KU13. procedures for producing test pieces for NDT, sample plates and production plates

KU14. default positions and datum points used by welding robots

KU15.
- procedures and good practice for writing welding robot programming with teach pendants including:-
- a. using the same wire stick out length
- b. ensuring axis moves of more than 180 degrees are made in two moves to ensure smooth
- motion
- c. robot posture is kept as close as possible to the home position
- d. using the wrist axis to position the torch angle
- e. avoiding excessive torch motion to create singularity on any axis
- f. avoiding multiple weld settings for the same joint type

KU16. procedures for trialling and editing welding robot programs

KU17. SOP recommended by the organisation for robotic welding

KU18. post-welding treatments

KU19. quality control and defect analysis of welded piece

KU20. various defects associated with the robotic process

KU21. various testing techniques like visual, destructive and non-destructive

KU22. safety requirements during the welding work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read work instructions, equipment manuals and process documents

GS2. communicate the process requirements to the supervisor and co-workers

GS3. attentively listen and comprehend the information given by the supervisor/team members

GS4. write work related information in English/regional language

GS5. recognise a workplace problem and take suitable action

GS6. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS7. plan and organise work according to the work requirements

GS8. complete the assigned tasks with minimum supervision

GS9. report to the supervisor or deal with a colleague individually, depending on the type of concern

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Prepare for robotic welding operations	**5**	**6**	**-**	**2**
PC1. identify the work to be done and product specifications by interpreting the engineering drawing,Welding Procedure Specification (WPS) and job orders	1	1	-	-
PC2. identify and arrange required tools, robotic arm, measuring instruments, accessories, consumables and input materials as per the requirements mentioned in WPS or drawing	3	1	-	1
PC3. check that material to be welded is as per the required specifications, quality and job requirement	-	1	-	-
PC4. plan the welding activities before starting the actual process as per WPS	-	1	-	-
PC5. set welding parameters like current, voltage, electrode size, material thickness, and joint type as per the SOP	1	1	-	1
PC6. follow appropriate safety practices as specified by organization during the work	-	1	-	-
Write robotic welding program	**15**	**26**	**-**	**12**
PC7. identify robotic technology, welding robot features and specifications including robot language and welding process and parameter capabilities to be used in welding applications	1	2	-	1
PC8. plan robot welding program	1	1	-	-
PC9. confirm required operations for robot arm and torch, types of weld, any positions or welds requiring non-standard instructions to be performed using appropriate technical reference source	1	1	-	-
PC10. review robot file library for correct references to weld size and weld type	1	1	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. identify suitable program code for robotic technology and welding applications	1	2	-	1
PC12. write joint arm and welding moves using teach pendant or software	1	2	-	1
PC13. use software including canned cycles and sub-routines for programming welding operations	1	2	-	1
PC14. program weld procedure for the robot using short, interrupted and continuous welds	1	2	-	1
PC15. create command instructions for welding instructions	1	2	-	1
PC16. perform any additional programming required including instructions, weld data recording and lower-level programming	1	2	-	1
PC17. check the sequence of the program as per the process sheet	1	1	-	-
PC18. test and debug the program for weld accuracy, compatibility and efficiency requirements by conducting trial run of the robotic arm on test material	1	2	-	1
PC19. verify trial welds and program performance against required specifications	1	2	-	1
PC20. edit program to adjust welding operations based on trial outcomes	1	2	-	1
PC21. save final program and complete operation sheets and any required records according to enterprise procedures	1	2	-	1
Perform robotic welding	**8**	**12**	**-**	**4**
PC22. install the work pieces and fixture on the apparatus and align them with the robotic arm as per the job requirements	1	1	-	-
PC23. set up robotic arm for functional operation in accordance with manufacturer's specifications	1	2	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. follow welding instructions, visualize weld paths, and check for potential defects during the welding process	-	1	-	-
PC25. perform welding by operating robotic arm and tack weld the joint at appropriate intervals in accordance with job plan, specifications, relevant welding standards and manufacturers'	2	3	-	1
PC26. monitor the welding process parameters (air pressure, electrode force, electrode distance, gas flow etc.) during the welding process	1	1	-	1
PC27. produce joints of the specified dimensional accuracy and required weld quality	1	2	-	1
PC28. measure the final welded piece and compare with the dimensions as prescribed in the WPS and engineering drawing	1	1	-	-
PC29. remove extra material, distortion etc. by using chipping hammers, grinders etc., from the welded piece	1	1	-	-
Perform post-welding operations	**2**	**6**	**-**	**2**
PC30. visually inspect welds to ensure that they meet specified standards and regulations	-	1	-	-
PC31. inspect welds to identify defects like porosity, undercuts, improper penetration etc	1	1	-	1
PC32. conduct destructive and non-destructive tests on the work pieces	1	2	-	1
PC33. tag and store the right quality pieces by following organizational policies and procedures	-	1	-	-
PC34. report to the supervisor about any problems faced or anticipated during the complete process	-	1	-	-
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N3134
NOS Name	Perform robotic welding operations
Sector	Automotive
Sub-Sector	
Occupation	Metal Joining
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

ASC/N3136: Perform welding and post welding operations

Description

This NOS unit is about performing all welding and post-welding operations as per the given work order and the standards specified by the organization.

Scope

The scope covers the following :

- Prepare for welding activities
- Perform welding operations
- Perform post-welding operations

Elements and Performance Criteria

Prepare for welding activities

To be competent, the user/individual on the job must be able to:

PC1. identify the final output product based on the engineering drawing, Welding Procedure Specification (WPS) and job orders

PC2. identify the tools, measuring instruments and input materials required for the job

PC3. select the appropriate welding method on the basis of drawing, WPS and job orders information

PC4. select and arrange the right material, equipment, fixtures and accessories as per the SOP and job requirements

PC5. check the input material, tools and equipment for any defects and that they are as per the required quality standards

PC6. fill CLRI (Clean, Lubricate, Retighten & Inspection) check sheet and report to the supervisor about any abnormalities identified and action taken to resolve them

PC7. set the welding machine and its parameters as per the selected welding method

PC8. install the work pieces and fixture on the apparatus and align them with the electrodes as per the job requirements

PC9. ensure that electrodes distance, contact area, pressure, application etc. are maintained as specified in Work Instructions (WI)

Perform welding operations

To be competent, the user/individual on the job must be able to:

PC10. start the welding machine for welding operations

PC11. perform welding process (SMAW/MIG/MAG/TIG/Robotic welding) as per SOP

PC12. ensure the welding process parameters (air pressure, electrode force, electrode distance, gas flow, etc.) are within standards by reading the various gauges and correct them if not within standards

PC13. support in comparing the dimensions of the final welded piece as prescribed in the work order and engineering drawing

PC14. check and repair the remaining material from the welded piece by using chipping hammers, grinders etc., as prescribed in SOP

PC15. check the hammered work piece to get the desired shape, if there are any welding bulges/distortions

Perform post-welding operations

To be competent, the user/individual on the job must be able to:

PC16. check the work pieces as per the work instructions for product quality

PC17. conduct destructive and non-destructive tests on the work pieces

PC18. check the issues identified in defective or to be repaired/reworked welded pieces and maintain a record of the same

PC19. check if Automotive Welding Machine Operator (Manual and Robotics) is tagged and stored the right quality pieces by following organisational policies and procedures

PC20. check the machine operations for any malfunctions/defects in the component

PC21. remove chips from different machine areas and dispose scrap or waste material in accordance with the company policies and environmental regulations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the basic principle of welding process

KU2. various types of welding such as SMAW, MIG, MAG, TIG, Resistance Welding (Seam Welding, Projection Welding), Robotic Welding etc. and their process flow

KU3. various types of welding joints

KU4. how to read and interpret welding drawings and symbols

KU5. SOP recommended by the manufacturer for using tools, measuring instruments, accessories etc. during the welding processes

KU6. ISO colour codes for welding apparatus such as gas cylinder, hoses, electric cables, etc.

KU7. different cleaning methods for electrodes, metal surfaces etc.

KU8. impact of various welding parameters like voltage, current, gas flow rate, speed, pressure, torch angle, cycle time, electrode distance etc. on the quality and quantity of welding

KU9. SOP recommended by the organisation for operating welding machine and its accessories

KU10. SOP recommended by the organisation for checking irregularities in the product/work piece

KU11. safety requirements during the welding work

KU12. the post welding processes like inspection, cleaning, maintenance etc.

KU13. various types of weld defects such as spatter, blow-hole, burn through, etc. and their remedies

KU14. methods of storage and tagging of final product

KU15. about the various testing techniques like visual, destructive and non-destructive

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret drawings, work instructions, equipment manuals and process documents

GS2. communicate the welding process requirements to the lead technician and co-workers

GS3. communicate issues to the supervisor that occur during welding process

GS4. attentively listen and comprehend the information given by the lead technician/team members

GS5. write any work related information in English/regional language

GS6. recognise a workplace problem and take suitable action

GS7. analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently

GS8. plan and organize tools, machines and consumables for carrying out welding job

GS9. complete the assigned tasks with minimum supervision

GS10. report to the supervisor or deal with a colleague individually, depending on the type of concern

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Prepare for welding activities	**10**	**16**	**-**	**7**
PC1. identify the fInal output product based on the engineering drawing, Welding Procedure Specification (WPS) and job orders	1	2	-	1
PC2. identify the tools, measuring instruments and input materials required for the job	2	1	-	2
PC3. select the appropriate welding method on the basis of drawing, WPS and job orders information	1	2	-	-
PC4. select and arrange the right material, equipment, fixtures and accessories as per the SOP and job requirements	1	1	-	-
PC5. check the input material, tools and equipment for any defects and that they are as per the required quality standards	1	2	-	1
PC6. fill CLRI (Clean, Lubricate, Retighten & Inspection) check sheet and report to the supervisor about any abnormalities identified and action taken to resolve them	1	2	-	-
PC7. set the welding machine and its parameters as per the selected welding method	1	2	-	2
PC8. install the work pieces and fixture on the apparatus and align them with the electrodes as per the job requirements	1	3	-	1
PC9. ensure that electrodes distance, contact area, pressure, application etc. are maintained as specified in Work Instructions (WI)	1	1	-	-
Perform welding operations	**11**	**18**	**-**	**6**
PC10. start the welding machine for welding operations	1	1	-	-
PC11. perform welding process (SMAW/MIG/MAG/TIG/Robotic welding) as per SOP	4	7	-	3

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure the welding process parameters (air pressure, electrode force, electrode distance, gas flow, etc.) are within standards by reading the various gauges and correct them if not within standards	2	3	-	2
PC13. support in comparing the dimensions of the final welded piece as prescribed in the work order and engineering drawing	2	3	-	1
PC14. check and repair the remaining material from the welded piece by using chipping hammers, grinders etc., as prescribed in SOP	1	2	-	-
PC15. check the hammered work piece to get the desired shape, if there are any welding bulges/distortions	1	2	-	-
Perform post-welding operations	**9**	**16**	**-**	**7**
PC16. check the work pieces as per the work instructions for product quality	2	2	-	1
PC17. conduct destructive and non-destructive tests on the work pieces	2	4	-	2
PC18. check the issues identified in defective or to be repaired/reworked welded pieces and maintain a record of the same	1	3	-	1
PC19. check if Automotive Welding Machine Operator (Manual and Robotics) is tagged and stored the right quality pieces by following organisational policies and procedures	1	2	-	-
PC20. check the machine operations for any malfunctions/defects in the component	2	3	-	2
PC21. remove chips from different machine areas and dispose scrap or waste material in accordance with the company policies and environmental regulations	1	2	-	1
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N3136
NOS Name	Perform welding and post welding operations
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Metal Joining
NSQF Level	4
Credits	6
Version	1.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).

4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.

5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take

subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9803.Organize work and resources (Manufacturing)	50	30	0	20	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	0	0	50	5
ASC/N9805.Interpret engineering drawing	50	30	0	20	100	10
ASC/N3134.Perform robotic welding operations	30	50	-	20	100	30
ASC/N3136.Perform welding and post welding operations	30	50	-	20	100	45
Total	**180**	**190**	**-**	**80**	**450**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
PPE	Personal Protective Equipment
PwD	Person with Disability
SOP	Standard Operating Procedure
GD&T	Geometric Dimensioning & Tolerancing
CAD	Computer-Aided Drafting
CAM	Computer-Aided Manufacturing
WPS	Welding Procedure Specification
CLRI	Clean, Lubricate, Retighten & Inspection
SMAW	Shield Metal Arc Welding
MIG	Metal Inert Gas
MAG	Metal Active Gas
TIG	Tungsten Arc Welding
ISO	International Organization for Standardization
SOP	Standard Operating Procedure
GD&T	Geometric Dimensioning & Tolerancing
CAD	Computer-Aided Drafting
CAM	Computer-Aided Manufacturing
WPS	Welding Procedure Specification
CLRI	Clean, Lubricate, Retighten & Inspection
SMAW	Shield Metal Arc Welding

MIG	Metal Inert Gas
MAG	Metal Active Gas
TIG	Tungsten Arc Welding
ISO	International Organization for Standardization
PPE	Personal Protective Equipment
PwD	Person with Disability
SOP	Standard Operating Practices
SOP	Standard Operating Procedure
GD&T	Geometric Dimensioning & Tolerancing
CAD	Computer-Aided Drafting
CAM	Computer-Aided Manufacturing
WPS	Welding Procedure Specification
CLRI	Clean, Lubricate, Retighten & Inspection
SMAW	Shield Metal Arc Welding
MIG	Metal Inert Gas
MAG	Metal Active Gas
TIG	Tungsten Arc Welding
ISO	International Organization for Standardization
WPS	Welding Procedure Specification
CLRI	Clean, Lubricate, Retighten & Inspection
SMAW	Shield Metal Arc Welding
MIG	Metal Inert Gas
MAG	Metal Active Gas
TIG	Tungsten Arc Welding
ISO	International Organization for Standardization
WPS	Welding Procedure Specification

CLRI	Clean, Lubricate, Retighten & Inspection
SMAW	Shield Metal Arc Welding
MIG	Metal Inert Gas
MAG	Metal Active Gas
TIG	Tungsten Arc Welding
ISO	International Organization for Standardization

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.